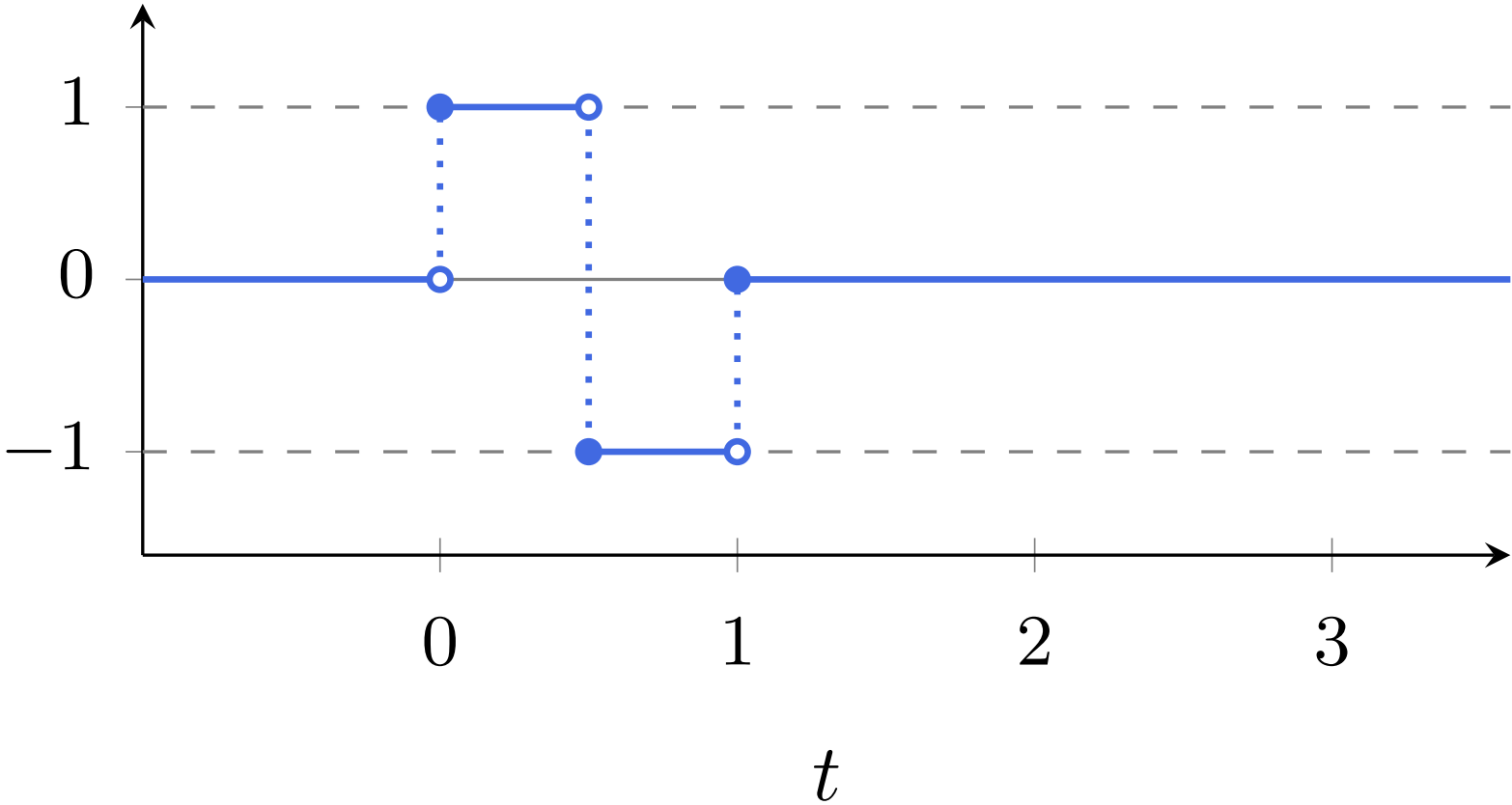(a) $\psi^{\mathrm{H}} = \psi^{\mathrm{H}}_{1,0}$